Phan Ngoc Hoa

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Education

- **Ho Chi Minh City University of Technology (HCMUT) - VNUHCM** Ho Chi Minh City, Viet Nam
  Pursuing a Bachelor of Computer Science *October 2022 – Present*
  - GPA: 3.2

Skills

- **Programming languages:** C, Python, Java
- **Embedded & IoT Frameworks:** ESP-IDF, FreeRTOS, LVGL
- **Hardware Protocols & Interfaces:** CAN Bus, OBD-II, I2C
- **Database Management Systems:** MySQL, MSSQL, PostgreSQL
- **Other Tools:** Git, GitHub, Figma, Postman, LaTeX, Docker

Projects

- **Automotive Head-Up Display (CHUD)** Personal Project | *April 2026 – June 2026*
  - **Description:** Designed and developed an embedded Head-Up Display for vehicles that reads real-time driving telemetries (Speed, RPM, battery voltage) from the ECU via OBD-II and renders them onto an LCD screen, enhancing driving safety and vehicle health monitoring.
  - **Technologies:** ESP-IDF, FreeRTOS, C, LVGL, TWAI/CAN Bus, OBD-II Protocol.
  - **Responsibilities:**
    * **Dual-Core Multitasking Architecture:** Architected a dual-core application using FreeRTOS, dedicated Core 1 for high-priority CAN Bus data polling (5Hz) to prevent timing jitter, and Core 0 for fluid 60 FPS LVGL UI rendering.
    * **ECU Integration:** Developed an OBD-II parser using CAN Bus driver to poll and decode real-time vehicle PIDs (Speed, engine RPM, coolant temperature, battery voltage).
    * **Low-Power & Standby Mode:** Programmed the ESP32 Ultra-Low Power RISC-V co-processor to monitor battery voltage, enabling automatic deep-sleep and wake-up cycles to prevent car battery drain.
    * **Hardware Calibration:** Integrated an ambient light sensor driver with ADC oneshot mode and hardware curve/line-fitting calibration schemes to adjust display backlight dynamically for optimal driver visibility.
  - **Github link:** https://github.com/ngochoaphan2004/CHUD.

- **Decentralized Federated Learning in Financial Systems** Graduation Project | *August 2025 – June 2026*
  - **Description:** Developed an innovative, privacy-preserving AI algorithm (DFedADP) optimized for detecting financial fraud across distributed networks with highly unbalanced . Additionally, engineered a full-stack banking application to simulate real-world transaction workflows.
  - **Technologies:** Python, Java Spring Boot, Next.js, P2PFL.
  - **Responsibilities:**
    * **Intelligent Weighting Mechanism:** Designed a smart scoring system that evaluates the reliability of AI updates from different distributed nodes, minimizing the impact of misleading or noisy data to ensure high model accuracy.
    * **Banking Simulation & System Integration:** Developed a full-stack banking simulation application adhering to the operational workflows of HSBC and OCB using Java Spring Boot and Next.js. This provided a functional environment to demonstrate transaction pipelines, and data exchange with the decentralized fraud detection system.
    * **Comprehensive Performance Validation:** When evaluated across diverse data scenarios, the algorithm effectively reduced client drift on the MNIST dataset, achieving 95.71% accuracy under IID conditions and maintaining a robust 80.55% under extreme Non-IID settings—outperforming standard decentralized methods by 39% to 69%. Furthermore, validation on a highly imbalanced simulated card transaction dataset secured an optimal F1-Score of $\approx$ 0.38 and a critical Recall of $\approx$ 0.6.
  - **Github link:** https://github.com/DATN-252.
  - **Content of the article :** https://tinyurl.com/DFedADP.

Honors and Awards

- **Second Place - Provincial Physics Contest, Dong Nai**